TrueRehab-RF: what's left before AEII

Team brief, 9 November 2026.

The kit works. Now we need real data, the real Pi and a rehearsed demo.

10 days left
AEII: 19–20 Nov, HKCEC

Where we are

Radar (AWR2944PEVM)	**DONE**	Flashed, streaming 60 fps with 0 lost frames. Sees curls as clear V-shapes at ~1 m.
sEMG (MyoWare)	**DONE**	1000 samples/s over Wi-Fi, 0 gaps. Never worn on an arm yet.
Recorder (all sensors, one clock)	**DONE**	Radar + sEMG + IMU + webcam pose into one session folder.
Dashboard (EN / TH / Chinese)	**DONE**	Live radar, sEMG charts, replays of model output.
AI model	**NO DATA**	Built and tested (240k params), but only trained on fake data.
Raspberry Pi set v0.3	**NOT ON PI**	Installer and software ready on the SSD. Never installed on the real Pi.
IMU on the arm	**BROKEN**	Wiring drops out when the arm moves (lost 36 of 72 s).

The one thing that matters most: we have never recorded a real person doing the full exercise with the radar *and* the sEMG worn at the same time. Until we do, the AI has nothing real to learn from, and every score on screen is simulated. **Getting real sessions recorded this week unlocks everything else.**

Must do before AEII

#	Task	Time	Who
1	**Set up the real Raspberry Pi** Flash the 64-GB SD card, copy the set from the SSD, run install.sh and selftest.py, then plug in the radar (power first, then USB) and check the live view. Steps: deploy/pi/README.md.	half day	
2	**Run the 25-minute Wi-Fi check on the Pi** Four 5-min recordings (A/B/C/D) with the MyoWare on the desk, then recorder/link_report.py. Tells us whether the ~1-s Wi-Fi pauses are under control.	1 h	
3	**Decide the network we bring to the venue** HKCEC Wi-Fi has a login page, which the sensor boards can't get past. Pick one: Pi as its own hotspot (free, untested) or a small travel router (~1,500 THB, most reliable). Test it at home for an hour.	2 h + buy?	
4	**Record the first real session** One volunteer: MyoWare on battery (USB unplugged!) with fresh pads on the biceps, radar ~1 m in front, webcam pose running. Protocol: 5 s still, press s + 3 shrugs, then curl sets with space at each rep. Then run check_session.py.	2 h	
5	**Record at least 3 different people, then train** The model needs ≥ 3 people to train and test fairly. Use codes (S01, S02...), never names. Then train.py --sessions.	1–2 days	
6	**Write and rehearse the demo** What we show live, what is replayed, and what we say honestly (see the box on page 2). Practise the setup from a bag: power, cables, network, dashboard on a phone.	half day	

Should do (makes the result stronger)

#	Task	Time	Who
1	**Fix the IMU wiring** Solder the joints or use a locking connector, tape the cable to the arm. Then redo the curls recording with it.	2 h	
2	**Test the finer radar setting on the board** 3.88 cm instead of 5.94 cm per step. Check 60 fps, 0 lost frames, temperature, and two metal objects ~5 cm apart. Only switch over if it passes.	1 h	
3	**Left/right check** Hold a metal tray ~1 m in front, then to the left, then to the right, 10 s each. Tells us which way the radar's angle axis runs.	30 min	

4	**Settle two model choices with real data** Keep the radar's own points as input or not (they hurt on fake data), and how many frames per rep (128 crops long reps).	after training	
5	**Ask a physiotherapist about our numbers** Shrug > 2 cm, lean > 10°, ROM < 80 %, fatigue = 20 % sEMG frequency drop: all placeholders today.	1 call	
6	**Volunteer consent** A simple written consent for healthy volunteers, especially if any video is kept.	1 h	

After AEII (say “next step” if judges ask)

#	Task	Time	Who
1	**Live AI scores on the Pi** Needs rep detection from the radar alone and a model trained on real people.		
2	**Muscle micro-vibration from radar** Our core research question. Phase data is already in the model input; whether it carries a fatigue signal is unproven.		
3	**Personal fatigue–compensation profile** Learn each user's pattern across many sessions.		

Suggested plan for the next 10 days

Tue 10 – Wed 11	Thu 12 – Sat 14	Sun 15 – Mon 16	Tue 17 – Wed 18	Thu 19 – Fri 20
Pi setup, Wi-Fi check, venue network	Fix IMU; record real sessions (3+ people)	Train, check results, pick model settings	Demo script, rehearse, pack the kit	AEII at HKCEC

What we can say at AEII

✓ The radar sees the room and the arm live, 60 times a second.
✓ We record radar, sEMG and video on one clock.
✓ The full AI pipeline runs end to end.

What we must NOT claim

✗ That the radar already measures fatigue.
✗ That scores on screen are real, unless trained on real people.
✗ Any medical or clinical result.

Kit rules (so nothing breaks)

- Radar: plug in power first, then USB.
- One program per sensor at a time (radar or MyoWare).
- MyoWare on a person = battery only, never USB.
- Its power switch must be ON, even on USB.
- Never flash the radar without asking first.
- Hot radar warning at 85 °C: stop and let it cool.
- Subject codes only. No names, no raw video.

Write your name in the **Who** column next to what you'll take, and tell the group chat. Stuck on something? Every task has step-by-step notes in the repo (deploy/pi/README.md, model/README.md, recorder/README.md).